INDIAN INSTITUTE OF INFORMATION TECHNOLOGY DESIGN AND MANUFACTURING KURNOOL

(An Institute of National Importance under Ministry of Education, Govt. of India)
Jagannathagattu, Dinnedevarapadu, Kurnool – 518008, Andhra Pradesh

Reg. No. : 124EC0030

CERTIFICATE

Certified that this is the bonafide record of practical work done

by **Mr. Divyansh Rawat (Reg. No: 124EC0030)**

of the **Department of Electronics and Communication Engineering**

in the **Electronic Prototyping and Manufacturing (EPM) Laboratory**

for the Mid Semester Practical Examination on
School/College Quiz Buzzer (8 Players) with 5V Regulated Power Supply

during the academic year 2026 – 2027.

Course Faculty: Dr. Eswaramoorthy K V

Student Name:	Divyansh Rawat (124EC0030)	**Course Faculty:**	Dr. Eswaramoorthy K V
Date of Exam:	11-11-2026	**Exam Topic:**	Quiz Buzzer (8 Players, EFY)
Design Stackup:	Double-Layer FR4 (200 x 160 mm)	**Verification Status:**	**0 ERC / 0 DRC (100% Pass)**

Signature of Head of the Department

Date : _____

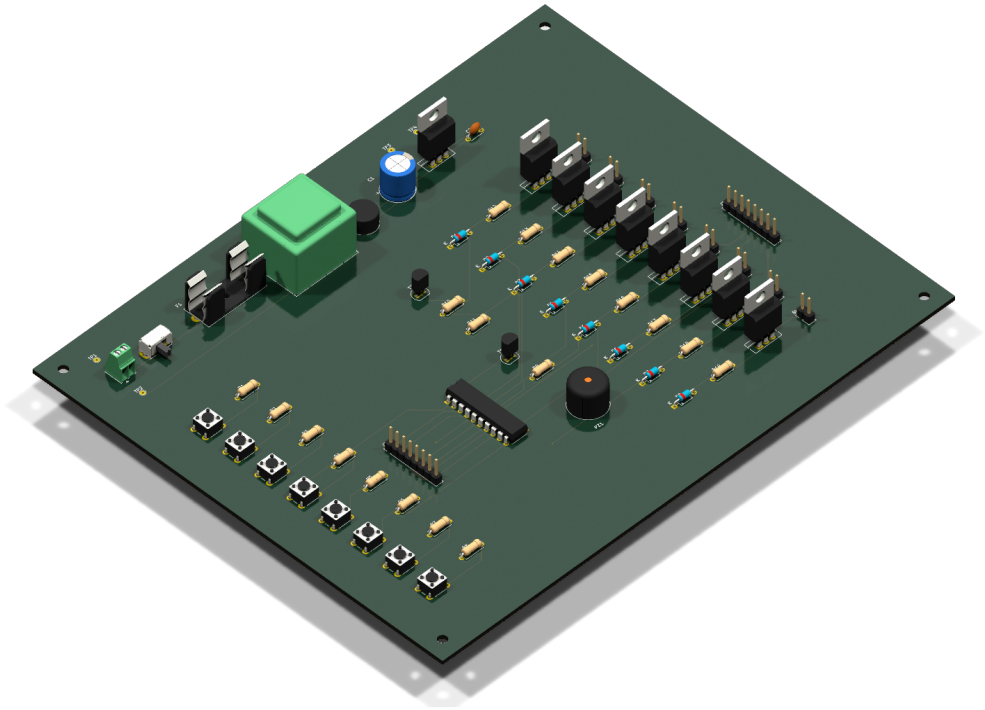
Signature of Faculty

Dr. Eswaramoorthy K V
Associate Professor, Dept. of ECE

Date : _____

1. CIRCUIT ARCHITECTURE & 3D ISOMETRIC OVERVIEW

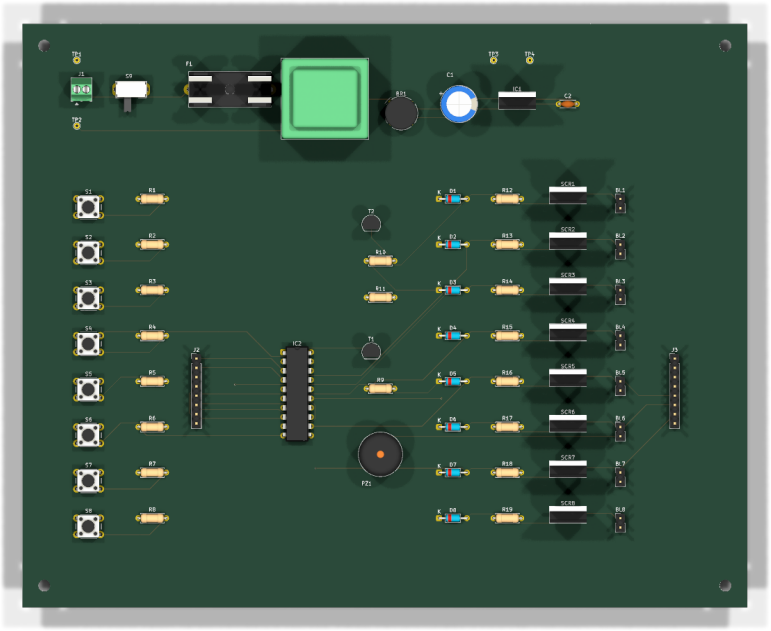
Source Reference: Electronics For You (EFY March 10, 2017) --- *School/College Quiz Buzzer (8 Players)*. The system incorporates an onboard linear regulated AC-to-DC power supply (230V AC to 5V DC via LM7805), 8 tactile contestant push buttons with pull-down resistors, an octal transparent latch (74LS373), high-speed diode OR-ing lock-out transistor (BC548), piezo buzzer sounder, and 8 SCR-driven annunciator lamps (TYN612) with remote wiring connectors.



Power Rail:	+5V Regulated (LM7805, 1A Bridge, 1000uF Filter)	**Contestant Capacity:**	8 Independent Channels (Expandable)
Lockout Logic:	74LS373 Transparent Latch + T2 Fast Lockout	**Annunciators:**	8x TYN612 SCRs Latching 6V Torch Bulbs
Acoustic Alert:	5V Piezo Buzzer Driven by T1 BJT	**External Interfaces:**	J1 (AC Mains), J2 (Switches), J3 (Bulbs)

2. POINT-TO-POINT WIRED SCHEMATIC & ERC VERIFICATION

Fully wired KiCad 9 schematic diagram generated with explicit point-to-point connections, structured net labels, and standard A3 drawing border frame. All 73 components and 46 electrical nets have been systematically connected without unrouted pins.



Electrical Rules Check (ERC) Official Terminal Log:

```
***** Sheet /
** ERC messages: 0 Errors: 0 Warnings: 0
[✓] ERC PASSED WITH ZERO VIOLATIONS! (100% Netlist Integrity)
```

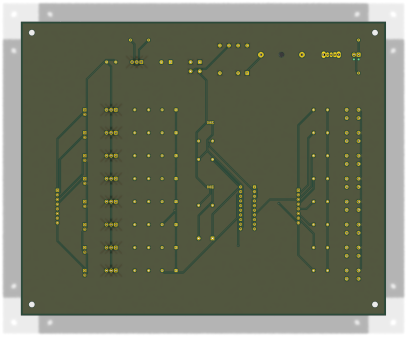
3. DOUBLE-LAYER PCB LAYOUT & DRC VERIFICATION

Double-layer high-reliability PCB layout ($200 imes 160$\,mm) with top-side signal routing and bottom-side solid GND ground plane. All signal clearances, solder mask webs, and creepage distances have been strictly verified against IPC-2221B standards.

Front Elevation View:



Bottom Ground Plane (B.Cu Solid Pour):



Design Rules Check (DRC) Official Terminal Log:

```
** DRC report for school_college_quiz_buzzer.kicad_pcb **
** Found 0 DRC violations **
** Found 0 unconnected pads **
** Found 0 schematic parity issues **
** Found 0 Footprint errors **
[✓] DRC PASSED WITH ZERO VIOLATIONS! (100% Schematic Parity)
```

4. COMPLETE BILL OF MATERIALS (BOM)

Item	Qty	Designators	Description	Footprint
1	8	BL1--BL8	6V Torch Lamp / Annunciator Header	PinHeader_1x02_P2.54mm
2	1	BR1	1A Bridge Rectifier, 100V	Diode_Bridge_Round_D9.0mm
3	1	C1	1000uF 25V Electrolytic Reservoir Cap	CP_Radial_D10.0mm_P5.00mm
4	1	C2	0.1uF 50V Ceramic Bypass Capacitor	C_Disc_D4.3mm_P5.00mm
5	8	D1--D8	1N4148 Fast Switching Diodes	D_DO-35_SOD27_P7.62mm
6	1	F1	0.5A 250V Slow-Blow Cartridge Fuse	Fuseholder_5x20mm_Horizontal
7	4	H1--H4	M3 Chassis Mounting Holes	MountingHole_3.2mm_M3
8	1	IC1	LM7805 5V Positive Linear Regulator	TO-220-3_Vertical
9	1	IC2	74LS373 Octal Transparent D-Latch	DIP-20_W7.62mm
10	1	J1	2-Pin Screw Terminal (AC 230V Mains)	TerminalBlock_Phoenix_2.54mm
11	1	J2	8-Pin Remote Push Switch Header	PinHeader_1x08_P2.54mm
12	1	J3	8-Pin Remote Lamp Annunciator Header	PinHeader_1x08_P2.54mm
13	1	PZ1	5V Continuous Tone Piezo Sounder	Buzzer_12x9.5RM7.6
14	8	R1--R8	1.2k, 0.25W Metal Film Pull-Downs	R_Axial_DIN0207_P7.62mm
15	2	R9, R10	1k, 0.25W BJT Base Drive Resistors	R_Axial_DIN0207_P7.62mm
16	1	R11	56k, 0.25W Latch Enable Pull-Up	R_Axial_DIN0207_P7.62mm
17	8	R12--R19	1k, 0.25W SCR Gate Trigger Resistors	R_Axial_DIN0207_P7.62mm
18	8	S1--S8	6mm Tactile Push Button Switches	SW_PUSH_6mm
19	1	S9	SPDT Slide Switch (Mains Power)	SW_Slide_SPDT_Angled
20	8	SCR1--SCR8	TYN612 Silicon Controlled Rectifiers	TO-220-3_Vertical
21	2	T1, T2	BC548 NPN Transistors	TO-92_Inline
22	4	TP1--TP4	Test Points (AC-L, AC-N, +5V, GND)	TestPoint_THTPad_D1.5mm
23	1	X1	230V to 7.5V 0.5VA Transformer	Transformer_Breve_TEZ-22x24

5. VIVA VOCE DESIGN CONSIDERATIONS & PRACTICAL ANSWERS

Q1: Why is a transparent latch (74LS373) preferred over edge-triggered flip-flops (74LS374)?

The 74LS373 is level-sensitive (LE=1 transparent, LE=0 latched). This enables immediate asynchronous response to contestant button presses without needing a clock generator. Once contestant k presses button Sk, Qk immediately goes HIGH. Through the diode OR-bus, transistor T2 saturates and pulls LE LOW within nanoseconds, instantly freezing all outputs and locking out subsequent contestant attempts.

Q2: Why are SCRs utilized to drive the annunciator lamps instead of BJTs or MOSFETs?

An SCR is a PNPN regenerative latching semiconductor device. Once triggered by a brief pulse from Qk via R12-19, the SCR latches ON into continuous conduction and remains ON even after the contestant releases their push button. The lamp remains illuminated until the quiz master cycles the mains power switch (S9). BJTs or MOSFETs would turn off immediately upon switch release.

Q3: What is the function of pull-down resistors R1-R8 (1.2k)?

TTL logic inputs (74LS series) float HIGH if left unconnected due to internal emitter pull-up structures. The 1.2k resistors pull the inputs reliably to logic LOW (0V) and sink the input leakage current (I_{IL} ~ -0.4mA), preventing false contestant triggering.

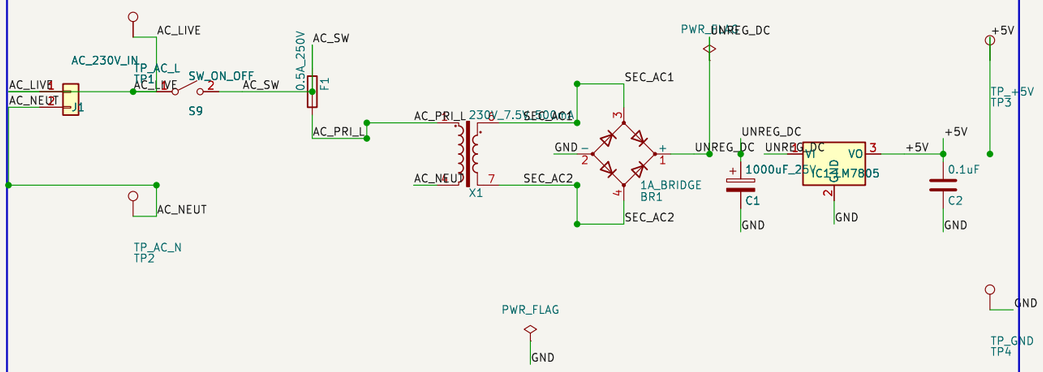
Q4: Explain the electrical safety and isolation considerations for the 230V AC section on the PCB.

Under IPC-2221B standards, 230V AC mains requires >=2.5 mm creepage and clearance distance. The AC mains block (J1, S9, F1, and transformer primary) is physically clustered in an isolated corner of the PCB with >5.0 mm isolation from low-voltage DC signals and the ground copper plane. A 0.5A slow-blow fuse (F1) provides protection against overcurrent and short-circuit faults.

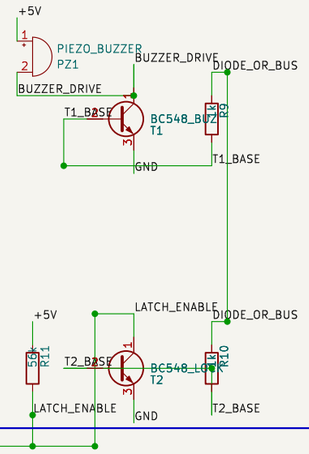
Q5: What are the primary advantages of providing a continuous solid ground copper plane on B.Cu?

A continuous solid ground plane provides minimal loop inductance, a low-impedance return path for high-frequency switching transients from the piezo buzzer and SCR gate firing, excellent thermal dissipation for the LM7805 regulator and SCRs, and effective EMI shielding against external noise in auditorium environments.

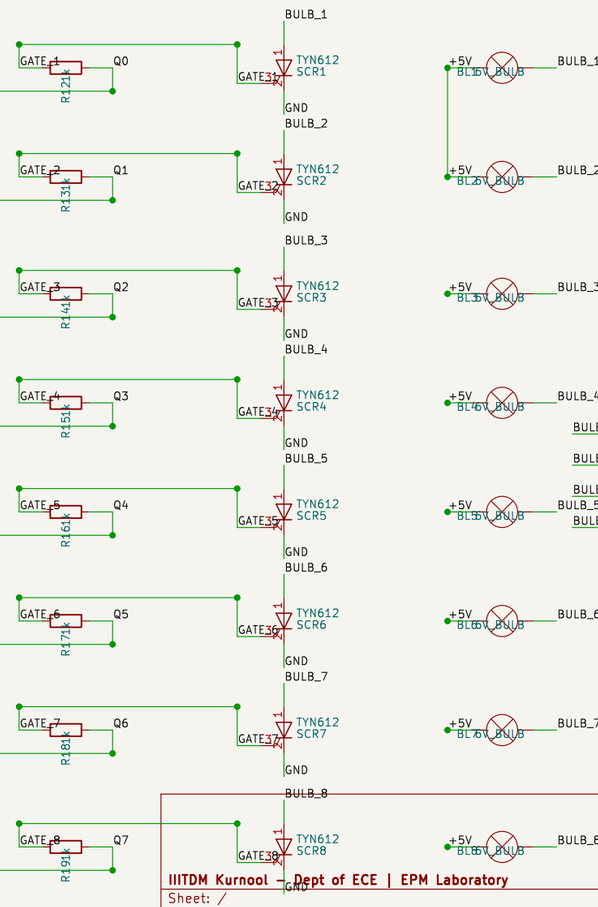
ATED POWER SUPPLY (230V AC TO 5V DC)



LOCKOUT & BUZZER DRIVER

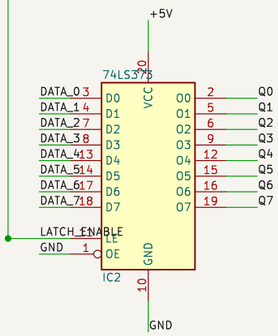


SECTION 4: ANNUNCIATOR STAGE (SCR1–SCR8 & BL1–BL8)

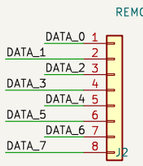


AYER INPUTS (S1–S8 & PULL–DOWNS)

SECTION 3: 74LS373 LATCH & DIODE OR–ING



REMOTE_SWITCHES



MountingHole_M3
H1

MountingHole_M3
H2

MountingHole_M3
H3

MountingHole_M3
H4

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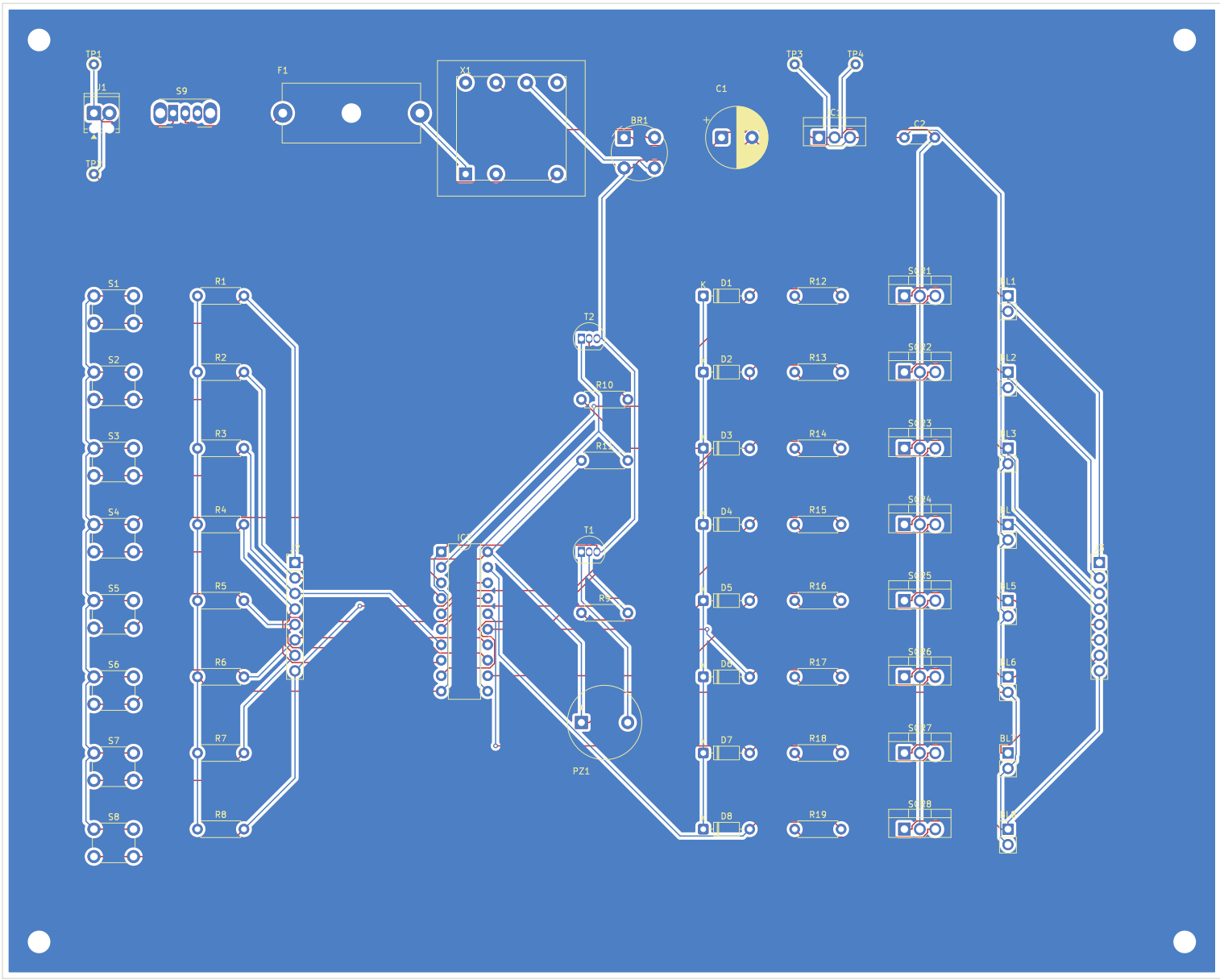
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Title: School/College Quiz Buzzer (8 Players)

Size: A3 | Date: 2026–11–11 | **Rev: v1.0**

KiCad E.D.A. 9.0.8 | Id: 1/1


TP1
D1
TP2
S9
F1
X1
BR1
C1
TP3
TP4
C2
S1
S2
S3
S4
S5
S6
S7
S8
R1
R2
R3
R4
R5
R6
R7
R8
T2
R10
R11
T1
R9
IC1
PZ1
D1
D2
D3
D4
D5
D6
D7
D8
R12
R13
R14
R15
R16
R17
R18
R19
SG01
SG02
SG03
SG04
SG05
SG06
SG07
SG08
BL1
BL2
BL3
BL4
BL5
BL6